

Week 9: Node.js + Express.js Basics

1. Introduction to Node.js

Node.js is a JavaScript runtime environment that allows developers to run JavaScript code outside the browser. It is mainly used for backend development and server-side applications. Node.js follows a non-blocking, event-driven architecture, which makes it highly efficient for handling multiple requests at the same time. It is widely used in modern web development because of its speed, scalability, and ability to build real-time applications like chat systems and APIs. Understanding Node.js is important for building strong backend systems in MERN stack development.

2. Express.js Framework

Express.js is a lightweight and flexible framework built on top of Node.js. It simplifies the process of creating web servers, APIs, and handling HTTP requests. Express provides features such as routing, middleware integration, and simplified server-side logic. It helps developers write cleaner and more manageable backend code. Because of its simplicity and efficiency, Express.js is one of the most commonly used frameworks for building scalable web applications and RESTful APIs.

3. Routes, Middleware, Request-Response Cycle

Routes are used to define how an application responds to a specific client request. For example, when a user visits a URL or sends a request, the route determines which function or response should be executed. Middleware functions are special functions that run between the request and response cycle. They can be used for tasks like logging, authentication, validation, and error handling. Understanding the request-response cycle is important because it explains how data flows from the client to the server and back to the client. This concept is essential for backend application logic.

4. REST APIs Basics

REST APIs (Representational State Transfer Application Programming Interfaces) are used for communication between frontend and backend applications. They allow clients to send requests and receive responses in a structured way, often using JSON data. Common HTTP methods include `GET` for fetching data, `POST` for creating data, `PUT` for updating records, and `DELETE` for removing data. REST APIs are widely used in MERN stack projects because they make data handling efficient, organized, and scalable for full-stack applications.

5. npm Packages & Dependencies

npm (Node Package Manager) is used in Node.js projects to install and manage external libraries or dependencies. Developers can install packages like Express, Nodemon, Axios, and many others to simplify development. Dependencies help save time and improve project functionality. Managing packages correctly is an important part of backend development.

Conclusion

This week focused on backend basics using Node.js and Express.js. By learning Node.js runtime, Express framework, routing, middleware, request-response cycle, REST APIs, and

npm package management, a strong backend foundation was developed. These concepts are essential for building scalable, secure, and efficient server-side applications in MERN stack development.